CH620003 Process Modelling and Simulation

Assignment-1

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Q1: NN CORRELATION OF BOILING POINT WITH CHEMICAL PROPERTIES.

Link to the data set (excel file):
https://www.dropbox.com/scl/fi/zlkxnyaptvc48xiu76xpy/data_file.xlsx?rlkey=754ozed754fyw4dhnlugjw7fa&dl=0

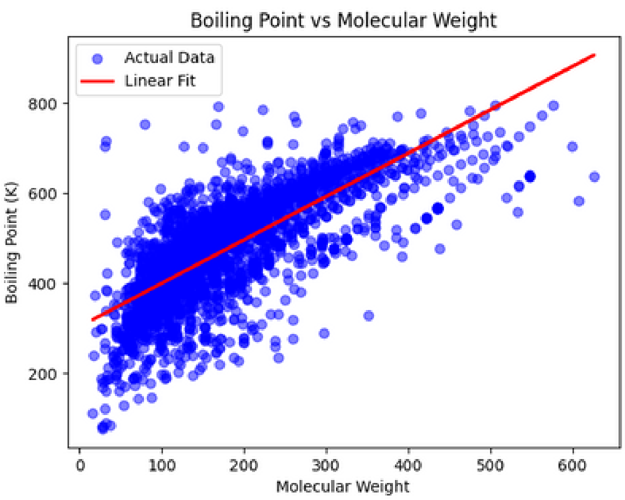
The data are tabulated into rows, one for each compound, and columns that include a common name, molecular weight, critical temperature (K), acentric factor, and the normal boiling point (K). Extract the data and place it into a matrix (array) AllData to use further on.

(a) Extract the boiling points from the data and place them in a column vector. Similarly, extract the molecular weights in a second vector. Plot the normal boiling point as a function of the molecular weight of the compounds in the database. Fit a straight line through the data and find the R^2 value.

Ans: coded the OLS, NN and Linear regression for this problem and providing the results here

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.580
Model:                            OLS   Adj. R-squared:                  0.580
Method:                 Least Squares   F-statistic:                     8325.
Date:                Sun, 09 Feb 2025   Prob (F-statistic):               0.00
Time:                        07:54:39   Log-Likelihood:                -33435.
No. Observations:                6031   AIC:                         6.687e+04
Df Residuals:                    6029   BIC:                         6.689e+04
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const        303.5846      1.894    160.328      0.000     299.873     307.297
x1             0.9640      0.011     91.241      0.000       0.943       0.985
==============================================================================
Omnibus:                      664.158   Durbin-Watson:                   1.068
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2569.792
Skew:                          -0.505   Prob(JB):                         0.00
Kurtosis:                       6.034   Cond. No.                         426.
==============================================================================
```

The **R2-score** for OLS: **0.58**
and the F-statistic value is 8325 suggesting that the model is highly significant and p value is almost 0.



(b) While we could use all the data we have to produce a meaningful correlation, let us assume that not all of the data were available. We will select a training set, or set of data that will be used to inform our ML algorithm. Extract from AllData 100 random data compounds. Using these selected compounds, build a 100 × 3 matrix, which we call X, where each row corresponds to a training example (i.e., the data for a given compound). For each one of these rows the first column is the number 1 (corresponding to x0) and the next two columns correspond to the values of x1 and x2 (molecular weight and acentric factor). Create another column matrix y with all the corresponding expected results; i.e., each of the elements of the vector is the expected reduced boiling point (Tb/ Tc) of each training example. The resulting $\boldsymbol{y}$ vector is a 100 × 1 vector.

The "solution" of this problem can be found directly as

$$\boldsymbol{\theta} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\,\boldsymbol{X}^T\,\boldsymbol{y}$$

Calculate the values of the coefficients θ_0, θ_1 and θ_2, to evaluate the quality of the correlation.

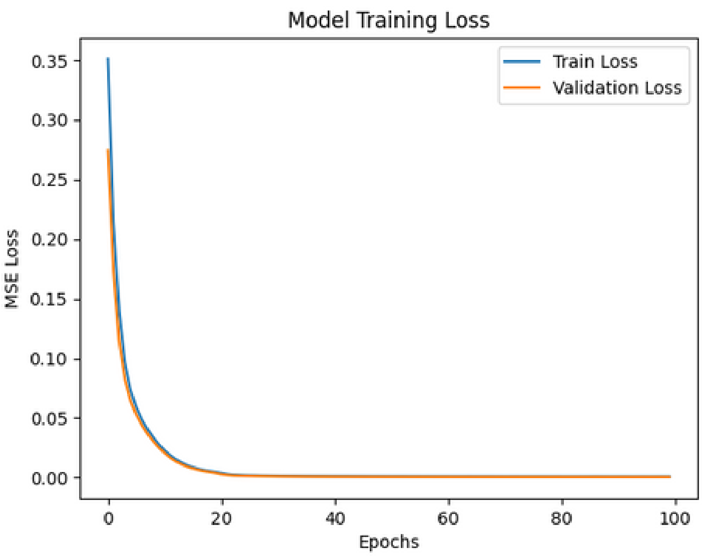
Ans: θ0, θ1 and θ2 respectively are: 5.95796567e-01 , 2.01872653e-04 and 1.54616024e-01

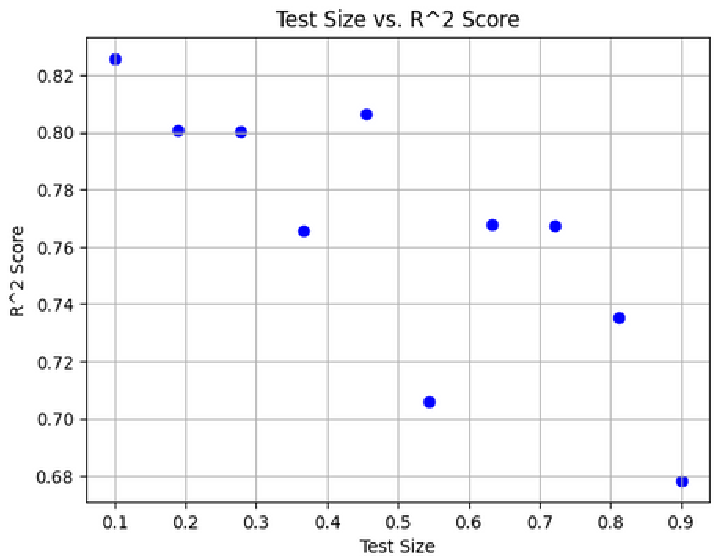
(c) From the matrix AllData retrieve two column vectors, one called InputData, which will have two columns, the MW and the acentric factor of all the data in the original set and a second matrix called TargetData which contains the reduced boiling point data.

Develop a neural network for this data, and train the network, using only 10% of the data. Present the results of the output.

You may want to explore the effect of changing the number of layers. In the above case, only a 10% of the available data is used to train the network (some 600 data points) while the rest is used for validations and testing. Explore what happens when these ratios are changed.

Ans: The **R2-score** for NN: **0.7478** although it can vary based on different splits.





Note: Source for the Antoine equation is given in the code

Q2: MULTICOMPONENT FLASH PROBLEM

A liquid mixture consisting of 100 kmol of 60 mol% benzene, 25 mol% toluene and 15 mol% o-xylene is flashed at 1 atm and 100 C. Assume ideal solutions, use vapour pressure data (from Antoine relations) to determine the K-values.

(a) Compute the component flow rates, mole fraction composition of the top and bottom products.

(b) Repeat the calculation in (a) for the same (1 atm) pressure, but different temperatures from 100 to 150 C. You need to think here before jumping on the calculations.

(c) Find the optimum temperature at which the recovery of benzene is maximum.

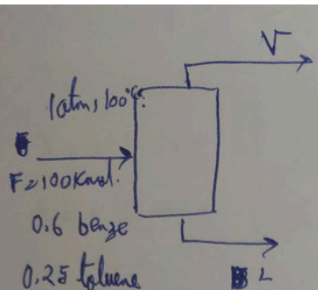
You need to do the numerical calculations, following the Rachford-Rice method of solution.

(d) If you recycle part of the bottom product stream, does the Benzene recovery increases? If yes, then how much?

Ans:

Q2

(a)



0.6 benze

0.25 toluene

0.15 o-xylene.

$V + L = F$

now we can write this as following:

$$x_b L + y_b V = 60$$

$$x_t L + y_t V = 25$$

$$x_o L + y_o V = 15$$

$$V + L = 100$$

$$y_i = k_i x_i \quad , \quad \Sigma y_i = 1$$

$$\log_{10} P_b = 6.90565 - \frac{1211.033}{320.79} \Rightarrow P_b = 1350.49 \text{ mm Hg.}$$

$$\log_{10} P_t = 6.95334 - \frac{1343.943}{319.377} \Rightarrow P_t = 556.32 \text{ mmHg.}$$

$$\log_{10} P_o = 6.99891 - \frac{1474.679}{313.69} \Rightarrow P_o = 198.56 \text{ mm Hg.}$$

So.

$K_b = 1.777$

$K_t = 0.732$

$K_o = 0.262$

$x_b V + y_b L = 60$

$V = (100 - L)$

$x_b V + k_b x_b (100 - L) = 60$

$x_b [L + 100 k_b - k_b L] = 60$

$x_b [177.7 - 0.777 L] = 60.$

$x_t [L + 73.2 - 0.732 L] = 25.$

$\therefore x_t [0.268 L + 73.2] = 25.$

and

$x_o [26.2 + 0.738 L] = 15.$

So.

$$\frac{60}{177.7 - 0.777 L} + \frac{25}{73.2 + 0.268 L} + \frac{15}{26.2 + 0.738 L} = 1$$

On solving

$\therefore L = 31.8273$ kmol

$V = 68.1727$ kmol.

So,

$x_b = 0.392$	$y_b = 0.696$
$x_t = 0.306$	$y_t = 0.224$
$x_o = 0.302$	$y_o = 0.080$

for top product;

benzene = ~~22.152 kmol~~ 47.448 kmol

toluene = ~~7.129 kmol~~ 15.270 kmol.

O-xylene = ~~2.546 kmol~~ 5.453 kmol.

bottom product:

benzene = ~~26.724 kmol~~ 12.476 kmol

toluene = ~~20.861 kmol~~ 9.739 kmol.

O-xylene = ~~20.588 kmol~~ 9.612 kmol.

(b) repeated the calculations in the code.

(c) $T_{opt} = 383K$

$= 110°C$

(d) on recycling the Bottom stream Benzene content will increase in the top product

as we will be considering a new Feed,

$F_1 = \cancel{F}^0 + L$

$z_b = \cancel{F}^0 z_{b,i} + L x_b$ (as not cont...

Q2

(d)

$F_1 = \cancel{F}^0 + L$

$F_1 = 31.8273$ Kmol

$\therefore z_b = 0.392$

$z_t = 0.306$

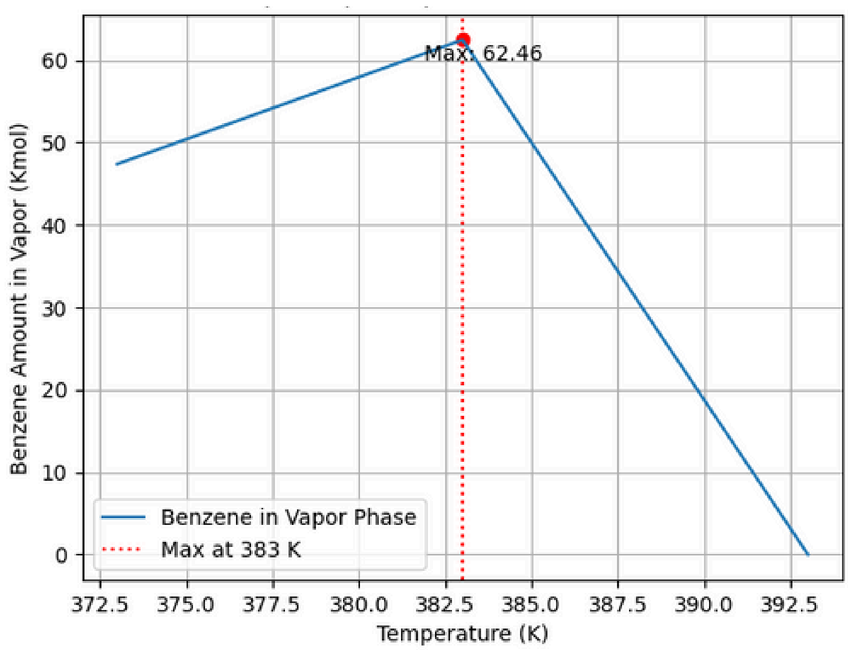
$z_o = 0.302$

On solving.

amount of Benzene recovered will be ≈ 0.

So recovery won't change and our initial assumption was wrong. @ 100°C,

After running the simulation for multiple Temperatures we found that the optimum temperature for getting maximum benzene recovery is **383K .**



Q3: BUBBLE POINT METHOD – MULTICOMPONENT MULTISTAGE DISTILLATION

1000 kmol/h of a saturated-liquid mixture of 60% methanol (normal boiling point 65 C), 20 mol% ethanol (normal boiling point 98 C), and 20 mol% n-propanol (normal boiling point 97 C) is fed to the middle stage of a distillation column having three equilibrium stages, a total condenser, a partial reboiler, all operated at 1 atm. The distillate rate is 600 kmol/h, and the external reflux rate is 2,000 kmol/h of saturated liquid.

(a) Performa ONE set of hand calculations for one iteration of the stage temperature.

(b) Write your own code to obtain converged solutions of the temperature profiles.

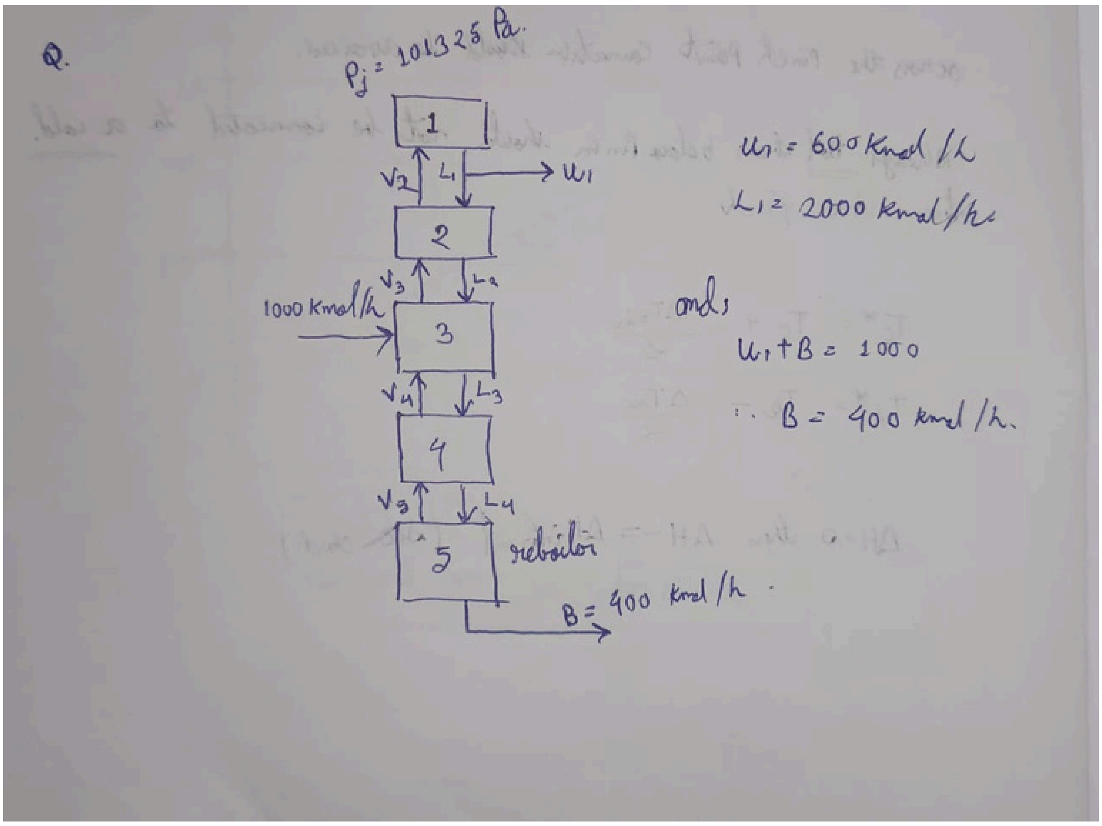
(c) Increase the number of equilibrium stages in the column to 10, and use the code (developed in part a) to determine the optimum feed location.

The K-values at 1 atm and the above temperatures for each component is expressed as $K = P_i^s / P$. The vapour pressure data are fitted to an extended Clausius-Clapeyron equation:

$P_s = \exp[C_1 + \frac{C_2}{T} + C_3 \ln T + C_4\, T^{C_5}]$ where P_s is in Pa and T is in K. The values of the constants $C_1 - C_5$ are presented in the following table:

Component		C1	C2	C3	C4	C5
M	Methanol	81.768	-6876	-8.7078	7.1926E-6	2
E	Ethanol	74.475	-7164.3	-7.327	3.1340E-6	2
P	Propanol	88.134	-8498.6	-9.0766	8.3303E-18	6

Ans:



1: methanol (0.6)
2: ethanol (0.2)
3: n-Propanol (0.2)

j	T	T_{sat} (K)	V	
1	65°C	338	0	actual Value
2	73.25°C	346.25	2600	
3	81.5°C	354.5	2600	assumed Value.
4	89.75°C	362.75	2600	
5	98°C	371	2600	

balancing for stage (1):

$\therefore V_2 = L_1 + U_1$

$\therefore V_2 = 2600$ kmol/h

K_{ij} values:

i \ j	1	2	3	4	5
1	1.0146	1.3907	1.8748	2.4890	3.2579
2	0.5793	0.8117	1.1258	1.5343	2.0575
3	0.2523	0.3698	0.5299	0.7436	1.0236

now calculating C_{ij} values.

$$C_{ij} = -V_{j+1} K_{i,j+1}$$

i \ j	1	2	3	4	5
1	-3615.9	-4874.6	-6471.4	-8470.8	0
2	-2110.4	-2927.1	-3989.3	-5349.5	0
3	-961.0	-1377.7	-1933.4	-2661.4	0

now Computing A_{ij} Values.

$$A_{i1} = 0$$

$$A_{i2} = -L_1 = -2000$$

$$A_{i3} = -L_2 = -2000$$

$$A_{i4} = -L_3 = -3000$$

$$A_{i5} = -L_4 = -3000$$

now, computing B_{ij} Values.

for this,

$$B_{ij} = (U_j + V_j) K_{ij} + (L_j + U_j)$$

$$L_j = V_{j+1} + \sum_{m=1}^{j} (F_m - U_m) \quad [U_j = 0]$$

$$\therefore B_{ij} = V_j K_{ij} + L_j + U_j$$

i \ j	1	2	3	4	5
1	2600	5615.9	7874.6	9471.4	8870.8
2	2600	4110.4	5927.1	6985.3	5749.5
3	2600	2961.5	4377.7	4933.4	3061.4

now the TDM for, methanol;

$$\begin{bmatrix} 2600 & -3615.9 & 0 & 0 & 0 \\ -2000 & 3615.9 & -4874.6 & 0 & 0 \\ 0 & -2000 & 7874.6 & -6471.4 & 0 \\ 0 & 0 & -3000 & 9471.4 & -8970.8 \\ 0 & 0 & 0 & -3000 & 8870.8 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{12} \\ x_{13} \\ x_{14} \\ x_{15} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 600 \\ 0 \\ 0 \end{bmatrix}$$

So,

$$\begin{bmatrix} x_{11} \\ x_{12} \\ x_{13} \\ x_{14} \\ x_{15} \end{bmatrix} = \begin{bmatrix} 0.9589 \\ 0.6895 \\ 0.4009 \\ 0.1821 \\ 0.0616 \end{bmatrix}$$

for ethanol:

$$\begin{bmatrix} 2600 & -2910.4 & 0 & 0 & 0 \\ -2000 & 4110.4 & -2927.1 & 0 & 0 \\ 0 & -2000 & 5927.1 & -33898 & 0 \\ 0 & 0 & -3000 & 63893 & -53495 \\ 0 & 0 & 0 & -3000 & 57495 \end{bmatrix} \begin{bmatrix} x_{21} \\ x_{22} \\ x_{23} \\ x_{24} \\ x_{25} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 200 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_{21} \\ x_{22} \\ x_{23} \\ x_{24} \\ x_{25} \end{bmatrix} = \cancel{\begin{bmatrix} 0.7935 \\ 0.8776 \\ 0.8306 \\ 0.5936 \\ 0.8097 \end{bmatrix}} \begin{bmatrix} 0.2645 \\ 0.32586 \\ 0.2769 \\ 0.1978 \\ 0.1032 \end{bmatrix}$$

for propanol;

$$\begin{bmatrix} 2600 & -361 & 0 & 0 & 0 \\ -2000 & 2691.5 & -1377.7 & 0 & 0 \\ 0 & -2000 & 4377.7 & -1933.4 & 0 \\ 0 & 0 & -3000 & 4933.4 & -2661.4 \\ 0 & 0 & 0 & -3000 & 3061.4 \end{bmatrix} \begin{bmatrix} x_{31} \\ x_{32} \\ x_{33} \\ x_{34} \\ x_{35} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 200 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_{31} \\ x_{32} \\ x_{33} \\ x_{34} \\ x_{35} \end{bmatrix} = \cancel{\begin{bmatrix} 0.3326 \\ 0.8970 \\ 1.2714 \\ 1.6399 \\ 1.6079 \end{bmatrix}} \begin{bmatrix} 0.0713 \\ 0.1927 \\ 0.3109 \\ 0.4011 \\ 0.3931 \end{bmatrix}$$

(b) Converged solution for the above is as following,

T1 = 339.7106306519236 K

T2 = 341.5880824407763 K,

T3 = 344.4698938177238 K,

T4 = 348.13815023913605 K,

T5 = 353.95260926186694 K

(c) If we use 10 stages instead of 3 the Feed should go to the 5th stage.
(Found this through simulation)(stage is counted excluding the reboiler and condenser here)